Smridhi Subedi

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EDUCATION

University of Texas at Dallas
B.S. Computer Engineering. *Expected Dec 2027*
Relevant Coursework: Embedded Design, Computer Architecture, Operating Systems, Data Structures & Algorithms, Machine Learning

AWARDS & HONORS

- Academic Excellence Scholarship
- Aspire2STEAM ASPIRER Scholarship (2024)

SKILLS

Embedded Systems & Firmware: MSP432P401R (ARM Cortex-M4F), TI SK-AM62, Arduino Uno, GPIO, ADC, UART, sensor interfacing, register-level programming, embedded debugging
Programming Languages & HDL: C, C++, Python, Java, JavaScript, TypeScript, Verilog, MIPS Assembly
Hardware Validation & CAD: Analog Discovery 2, oscilloscope, PSpice/OrCAD, MATLAB, AutoCAD, SolidWorks
Development Tools & Web: Code Composer Studio, Git/GitHub, VS Code, pytest, React, Tailwind CSS, Figma

PROJECTS

UTD HackEdge 2026 — SAAM / SAAMOS 2026 – Present
Team Lead & Audio Software Developer
- Lead a four-person team building SAAMOS, an offline assistive system for blind and low-vision users combining voice interaction, computer vision, and directional audio feedback on the TI SK-AM62 embedded Linux platform.
- Developed a Python text-to-speech module with automated pytest unit testing using mocked system commands for integration with locally processed voice commands and computer-vision output.

MSP432 Intrusion Alarm Controller
Embedded C / ARM Cortex-M4F
- Developed a three-state intrusion alarm on the MSP432P401R using register-level GPIO, pull-up resistors, active-low input handling, LED outputs, and latched alarm-state logic in embedded C.
- Diagnosed switch bouncing issues by refining firmware control logic; hardware-tested armed, disarmed, and triggered states to verify correct state transitions.

Analog Circuit Characterization & Debugging
Electronics Laboratory
- Construct and characterize analog circuits including diode rectifiers, RC/RL/RLC filters, BJT/JFET amplifiers, and MOSFET/CMOS circuits; measure voltage, gain, and frequency response using Analog Discovery 2 and compare against PSpice/OrCAD simulations.

WEHack 2025 — GrowYourTime 2025
Product Designer & Front-End Developer
- Built Figma prototypes and developed React, TypeScript, and Tailwind CSS components for a productivity web app; deployed via Vite to GitHub Pages.

EXPERIENCE

Teach/Tech Lead Mar 2023 – Jul 2024
iCode
- Led project-based instruction in VEX Robotics, drones, and IoT over 16 months for groups of up to 20 students; taught Python, Java, electronics, and robotics principles.

Sales Associate Jul 2024 – Oct 2024
Micro Center
- Advised customers on PC component selection, compatibility, and hardware issues; communicated technical tradeoffs across desktop systems, 3D printers, and electronics.

Website Developer Jun 2024 – Aug 2024
Aspire2STEAM
- Built and maintained WordPress components using HTML, CSS, and JavaScript; updated page structure, styling, and cross-device layouts.

ACTIVITIES & LEADERSHIP

Comet Robotics — Member | IEEE — Member | Nepali Student Association — Social Media Manager (2025)